ImTech Unfolded

A handheld Web AR flyer for the HSLU Bachelor in Immersive Technologies. Point a phone at the printed flyer and the study programme unfolds in 3D on top of it. No app download is needed.

TYPE	TRACKING	DEVICES	TEAM
Handheld Web AR	**Image target**	**iOS & Android**	**Solo**

A Project idea

The Head of Program and the Marketing department want handheld AR for their campaign about the Immersive Technologies bachelor. ImTech Unfolded is a small, polished version of that idea: **one printed A5 flyer and one URL/QR code** that turn a flat piece of paper into a 3D experience explaining the programme.

When the flyer is recognised, a floating XR headset appears above it with three hotspots around it. Each hotspot stands for one part of the programme: **Virtual Reality**, **Augmented Reality** and **3D & Game Development**. Tapping a hotspot opens a short info card. A clear call to action at the end links to the official programme page. The aim is a small, finished experience that feels production-ready, not a large one with loose ends. It will also be a portfolio piece that anyone can try with only a phone.

1 · Scan
QR on flyer opens the web app

2 · Load
Branded loading screen & camera permission

3 · Track
Flyer recognised, 3D scene appears

4 · Explore
Tap 3 hotspots to read info cards

5 · Act
“Discover the programme” link

B Applied tech stack

AREA	TOOL	WHY
XR framework	A-Frame 1.x + MindAR (image tracking)	Covered in SW02–SW04; open source (MIT); runs in Safari and Chrome with one build
Build & dev	Vite, vanilla JS (ES modules)	Fast local dev with HTTPS for phone testing; small production bundle
3D assets	Blender → glTF/GLB, gltf-transform (Draco + WebP textures)	Applies SW05 “3D models for the web”: compressed, small files
Target image	Flyer designed in Figma, compiled to `.mind` with the MindAR compiler	High-contrast, feature-rich target that tracks well (SW04)
Version control	GitLab: `main` + one `feature/*` branch per feature, merged by merge request	Clean, readable commit history as required
Hosting	HTTPS static hosting as taught in SW08 (fallback: Netlify)	Camera access requires HTTPS
Testing	iPhone (Safari), Android (Chrome), desktop Chrome DevTools	Weekly test cycle on real devices, plus friends testing without an explanation

A+ Visual concept: the experience on the phone

Everything runs in the phone's browser. The user scans the QR code on the flyer, points the camera at it and explores the 3D scene by tapping. They can walk around the flyer to see the scene from every angle. *These mockups are concept sketches; final styling will follow them.*



1 Start

Branded start screen with loading progress. Explains why the camera is needed before asking.



2 Scan

Camera view with a scan frame and hint until the flyer is recognised.



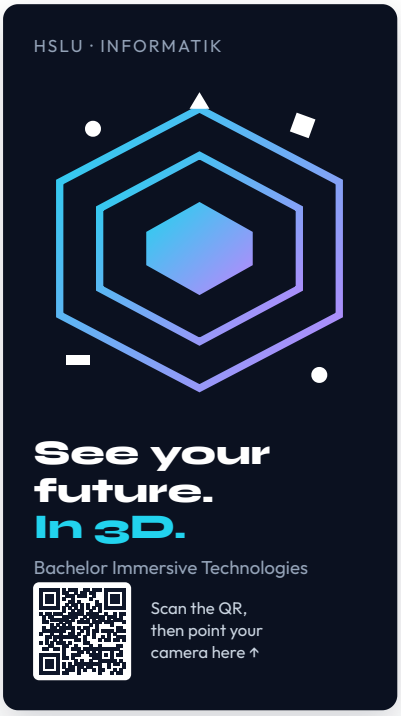
3 Explore

The headset floats above the flyer. Three tappable hotspots and a progress counter.



4 Learn & act

Info card slides up. A call to action links to the programme page.



The printed flyer (A5): the AR target

The flyer is both a normal print ad and the image target. The central graphic is built to track well: **high contrast, sharp edges, asymmetric details** and no large empty areas (SW04 “What makes a good target image”).

Interactions

Tap	Start AR · open hotspot · close card · CTA button
Move phone	View the 3D scene from any angle around the flyer
Move flyer	Scene follows the paper in real time
Lose flyer	Scene hides, “Point your camera…” hint returns
No camera	Same scene in the browser, drag to orbit (fallback)

c Feature specification & acceptance criteria

The features follow the user's path through the app. The usability measures from SW06 are built into the features they belong to (overview on page 5).

Feature 1 · First impression & loading screen

CORE

Something intentional is visible right away. The link looks good when shared, and loading never shows a blank page.

- Custom `<title>`, favicon and Open Graph tags (og:title, og:image) set
- Shared link shows the preview image + title in WhatsApp/Teams
- Logo + title visible ≤ 1 s after opening (no blank screen)
- Loading bar with text ("Loading 3D model… 72 %") from `a-asset-item` progress
- Loader hides the page until CSS and fonts are ready (no FOUC / FOUT)
- Loader disappears when `<a-scene>` fires `loaded`

Feature 2 · Start, permissions & error handling

CORE

The app checks what the device can do before it offers AR, explains why it needs the camera, and handles every failure with a readable message.

- Feature detection on load: HTTPS, camera (getUserMedia), WebGL
- "Start AR" shown only if all checks pass; otherwise the fallback (F3) is offered
- Start screen says in one sentence why the camera is needed
- Camera only starts after the tap on "Start AR"
- Camera denied → human message + how to re-enable it in browser settings
- No internet / model fails to load → message + "Reload" button
- Failure paths tested: permission denied, airplane mode, broken model file name
- No path ends on a black or blank screen

Feature 3 · Fallback: 3D view without AR

CORE

If AR can't run (no camera, denied, desktop browser), the user sees the same scene in the normal browser window, so it is never a dead end.

- Fallback reuses the same 3D scene, hotspots and info cards
- Touch drag (phone) / mouse drag (desktop) orbits around the model
- Reachable from the error message and from a "No flyer? View in 3D" link
- Works on desktop Chrome, Safari and Firefox without a camera

Feature 4 · Track the flyer & guide the user

CORE

The app recognises the printed flyer and anchors the 3D scene to it. Users are told what to do while searching and when tracking is lost.

- Flyer recognised within 2 s when held in front of the camera
- 3D scene sits centred on the flyer and follows it when moved
- Tracking works at a distance of 30–60 cm
- Runs at ≥ 30 FPS on a mid-range phone (tested on 2 devices)
- Hint "Point your camera at the flyer" shown until found, hidden when found
- Tracking lost > 1 s → scene hides + "Flyer lost, point at it again"
- MindAR's default scanning/loading/error UI replaced with own styled elements
- Jitter reduced via MindAR smoothing (filterMinCF / filterBeta tuned)

Feature 5 · Interactive hotspots & info cards

CORE

Three hotspots around the headset (VR, AR, 3D/Games) open short info cards. Everything that can be tapped looks tappable and reacts when tapped.

- 3 hotspots visible and gently pulsing to invite tapping
- Hint “Tap a hotspot to explore” until the first tap
- Tap feedback within 100 ms: scale + colour change (+ short vibration where supported)
- Card opens within 300 ms; only one card open at a time
- Cards are an HTML overlay above the canvas, not text inside the 3D scene
- Each card: title, ≤ 40 words, one icon; closes with ✕ or a tap outside
- Every touch target ≥ 44 × 44 px; body text ≥ 16 px, WCAG AA contrast
- Progress shown (“2 / 3 explored”); visited hotspots marked

Feature 6 · End state, legal & production polish

CORE

A clear next step at the end, a legal footer, and the technical polish that makes it feel like a finished web app.

- After 3 / 3: “Discover the programme” CTA (new tab) + “Explore again”
- Small Impressum · Privacy link on every screen (start, scanning, AR, fallback)
- Privacy note: camera images stay on the device; no analytics, no cookies
- Viewport meta set; no accidental page scroll or pinch-zoom during AR
- Screen stays on during AR (Wake Lock API where supported)
- All GLB models Draco-compressed, ≤ 2 MB total; initial load ≤ 5 MB
- Card icons and the end-state assets load deferred, after the target is found
- Build output hashed by Vite; files in `public/` renamed on update (no stale cache)
- One palette + one font pair as CSS variables (`--main-color`, `--font`); same button style everywhere
- Hosted on HTTPS and reachable via the QR code on the flyer

Long shots · if time allows

STRETCH

- **Multiple targets:** 3 poster variants (VR / AR / Games), each showing its own 3D model; switching in < 500 ms.
- **DE / EN toggle:** all UI text and cards switch language without a reload; the choice is remembered.
- **Intro animation & sound:** the headset “unfolds” out of the flyer (≤ 2 s) with an optional subtle sound (muted by default, started by the “Start AR” tap).

How I plan to use AI

- **Coding assistant (Claude):** I give it one feature at a time, together with that feature's acceptance criteria from this proposal as the definition of “done”.
- **Transparency:** every commit that contains AI-generated code is tagged `[AI]` in the commit message, with a short note on what was generated.
- **Debugging & review:** explaining errors, tuning MindAR parameters, checking for performance issues.
- **No one-shotting:** I review and test every AI suggestion on a real device before merging.
- **Not AI:** concept, flyer design, 3D scene composition and final card texts are done by me.

E Usability checklist (SW06) → where it is covered

SW06 CHECKLIST ITEM	COVERED IN	HOW
Something intentional visible right after opening	F1	Logo + title within 1 s, before any 3D loads
Loading shows progress or text	F1	Progress bar with “Loading 3D model… %”
Start button explains why the camera is needed	F2	One-sentence explanation on the start screen
Hint on how to start (what to point at)	F4	“Point your camera at the flyer” + scan frame
Message when tracking is lost	F4	“Flyer lost, point at it again” after 1 s
Clear message when camera denied / not supported	F2	Feature detection + human error messages
Fallback so there is no dead end	F3	Same scene in the browser, orbit by touch / mouse
Big model loaded only when needed	F1 · F6	Core model in `<a-assets>`; the rest deferred
Text and buttons readable on the real device	F5	≥ 16 px text, ≥ 44 px targets, HTML overlay
Title, icon, share preview	F1	`<title>`, favicon, og:title / og:image
One look everywhere	F4 · F6	CSS variables, one button style, restyled MindAR UI
Tested on real devices, bad network, permissions denied	F2 · Testing	iPhone + Android, airplane mode, denied permission

F Milestones

28.10	Proposal feedback · GitLab repo & Vite setup
04.11	F1 first impression & loader · F2 start & errors
18.11	F4 flyer tracking + first 3D scene
25.11	F5 hotspots & info cards
02.12	F3 fallback · F6 end state & polish
09.12	Hosting, failure-path & device testing
16.12	Demo station & oral report

G Risks & mitigation

Unstable tracking	High-contrast flyer; check its feature points in the MindAR compiler before printing
iOS Safari quirks	Test on a real iPhone from the first tracking build on
Slow mobile data	Compress early; defer everything not needed for the first view
Running out of time	Long shots only start once F1–F6 are done